

RL on Cartridges

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Abstract

RL on Cartridges: Freezes the base model and applies reinforcement learning directly to a small customer- or task-specific memory module, such as a learned KV-cache cartridge, so it acquires specialized agent skills without modifying the full model.

The high level idea is to apply *online* RLVR on top of the cartridge itself rather than the base policy. This should enable quasi-non-parametric adaption to new tasks, ability to learn from agent traces/environment feedback, and continually adapt to new tasks without forgetting.

We introduce **CARL** (chosen backronym): **Cartridge Adaptation through Reinforcement Learning**.

Introduction and Motivation

We want to learn from agent traces and environment feedback, and continually adapt to new tasks.

Enterprise relevance:

1. Customer-specific adaption

- Instead of $\theta \rightarrow \theta'$, you get $\theta + C_{\text{customer}}$, where every enterprise has its own evolving cartridge.

Base Model (shared)

+

Customer A cartridge

Customer B cartridge

Customer C cartridge

2. Cheaper RL

- Base model = 70B params
- Cartridge = 40M params
- RL scales roughly with the number of trainable parameters and optimizer state. Updating the cartridge only is dramatically cheaper than updating the base model.

3. Fast continual learning

- Changing a cartridge can happen every hour upon the presence of new user data.

4. Isolation

- Updating the model for one customer doesn't mess up the other customer
- A bad RL update only damages one customer's cartridge. It doesn't catastrophically interfere with everyone else.

5. Many parallel variants during inference

- Imagine OpenAI has 100,000 enterprises Each has different preferences. You cannot realistically keep 100,000 full models But you can keep 100,000 cartridges.

Goals and research contributions:

- Enable non-parametric¹ adaption to new tasks
- Learn from environment feedback/agent traces
- Learn OOD tasks from context/information
- Continually adapt to new tasks without forgetting

Related work

Cover:

- PEFT/LoRA work
 - A key question of concern here is whether this idea is truly different than just doing LoRA
 - (Asawa et al. 2025), (Asawa, Dimakis, and Zaharia 2026)
- Context distillation
- Cartridges (Eyuboglu et al. 2025).
 - SFT vs. RL analogy ² You can think of it like SFT + RL, where ordinary cartridges is like the base model that went through SFT and then CARL is like RLVR on top of the that (mid-training vs. post-training).

¹By non-parametric here, we mean that the core base model policy isn't being updated... just the cartridge itself.

²You can think of it like SFT + RL, where ordinary cartridges is like the base model

- Self-distillation

Method

Research question

We’d want to test things like:

- **RQ1 (RL on Cartridges)**: can we improve on a task with just RL’ing the cartridge without modifying the base model itself?
- **RQ2 (OOD Generalization)**: can we add OOD information/context/hints[$\hat{\text{pope}}$] into the cartridge, that doesn’t exist in the base policy, to then reinforce during RLVR? Thereby expanding support/pass@k?
- **RQ3: (Continual Learning)**: can we sequentially train on new tasks via new cartridges without degrading performance or inducing catastrophic forgetting?
 - Could also be good to analyze the weight space geometry/sparsity of the updates (i.e., SFT forgets, RL doesn’t)
- **RQ4 (Self-distillation)**: Cartridge OPSD: Use on-policy self-distillation to continually compile privileged experience into a trainable cartridge while keeping the base model frozen. Compare with RL optimizer.

[$\hat{\text{pope}}$] (for hints, see (Qu et al. 2026))

Experiments

Datasets and benchmarks

- Continual Learning Bench (Asawa et al. 2026)

Results and discussion

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Asawa, Parth, Alexandros G Dimakis, and Matei Zaharia. 2026. “SIEVE: Sample-Efficient Parametric Learning from Natural Language.” *arXiv Preprint arXiv:2604.02339*.

Asawa, Parth, Christopher M Glaze, Gabriel Orlanski, Ramya Ramakrishnan, Benji Xu, Asim Biswal, Vincent Sunn Chen, Frederic Sala, Matei Zaharia, and Joseph E Gonzalez. 2026. “Continual Learning Bench: Evaluating Frontier AI Systems in Real-World Stateful Environments.” *arXiv Preprint arXiv:2606.05661*.

- Asawa, Parth, Alan Zhu, Abigail O’Neill, Matei Zaharia, Alexandros G Dimakis, and Joseph E Gonzalez. 2025. “How to Train Your Advisor: Steering Black-Box Llms with Advisor Models.” *arXiv Preprint arXiv:2510.02453*.
- Eyuboglu, Sabri, Ryan Ehrlich, Simran Arora, Neel Guha, Dylan Zinsley, Emily Liu, Will Tennien, et al. 2025. “Cartridges: Lightweight and General-Purpose Long Context Representations via Self-Study.” *arXiv Preprint arXiv:2506.06266*.
- Qu, Yuxiao, Amrith Setlur, Virginia Smith, Ruslan Salakhutdinov, and Aviral Kumar. 2026. “POPE: Learning to Reason on Hard Problems via Privileged on-Policy Exploration.” *arXiv Preprint arXiv:2601.18779*.